

Lucas Toshio Ito

Biomedical scientist · PhD candidate

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São Paulo, Brazil

FOCUS	Psychiatric genetics · Suicidality · Polygenic scores · CNVs · Neuroimaging · Bioinformatics
PROFILES	Google Scholar · PubMed · ORCID · GitHub · LinkedIn
LANGUAGES	Portuguese (native) · English (advanced)

RESEARCH PROFILE

Biomedical researcher specializing in psychiatric genetics, with experience in genomics, neuroimaging, polygenic scoring, copy-number variant analysis, and bioinformatics. I am a PhD candidate at the Federal University of São Paulo (UNIFESP) under the supervision of Marcos Leite Santoro and completed a one-year visiting research period at the Icahn School of Medicine at Mount Sinai under the supervision of Niamh Mullins. I serve as a lead genetics analyst for the Brazilian High-Risk Cohort for Mental Health Conditions, integrating genomic, neuroimaging, and psychiatric data. My work emphasizes psychiatric genetics in underrepresented and admixed populations and the study of suicidal thoughts and behaviors across biological levels.

EDUCATION

03/2023–present	PhD, Structural and Functional Biology Psychiatric Genetics Federal University of São Paulo (UNIFESP), São Paulo, Brazil Supervisor: Marcos Leite Santoro Project: Impact of genetic variants on structural neuroimaging: a longitudinal approach in psychiatric disorders.
01/2025–01/2026	Visiting PhD Research Period · Icahn School of Medicine at Mount Sinai New York, United States · Supervisor: Niamh Mullins Project: The role of copy-number variants in suicide attempt and suicidal ideation in the Psychiatric Genomics Consortium. FAPESP-supported international research period completed as part of the UNIFESP PhD training.
03/2018–02/2023	BSc, Biomedical Sciences Federal University of São Paulo (UNIFESP), São Paulo, Brazil Undergraduate project: Polygenic scores and their relationship with multiple mental disorders in the Brazilian population.

RESEARCH EXPERIENCE

03/2023–present	PhD Researcher and Genetics Analyst · UNIFESP / BHRC Lead and contribute to genomic analyses in the Brazilian High-Risk Cohort for Mental Health Conditions. Integrate polygenic scores, common and rare variation, longitudinal phenotypes, cognition, and structural and functional neuroimaging. Develop collaborative projects on suicidality using CNVs, miRNA, neuroimaging, and genetic data.
01/2022–02/2023	Undergraduate Researcher · Psychiatric Genetics, UNIFESP Supervisor: Marcos Leite Santoro Analyzed polygenic scores and multiple psychiatric outcomes in a longitudinal Brazilian cohort of children and adolescents.
03/2022–12/2022	Practical Genetics Intern · Department of Morphology and Genetics, UNIFESP Training in karyotyping, arrays, MLPA, FISH, whole-exome analysis, nucleic-acid extraction, cDNA synthesis, Sanger sequencing, real-time PCR, digital PCR, and liquid biopsy.
06/2018–12/2021	Undergraduate Researcher · Proteomics and Peptidomics, UNIFESP Supervisor: Alexandre Keiji Tashima Proteomic and peptidomic characterization of Nephilingis cruentata spider and Lachesis muta snake venoms using mass spectrometry.

AWARDS AND HONORS

2025	Peter McGuffin Scholarship, World Congress of Psychiatric Genetics.
2025	Reviewers' Choice Abstract, American Society of Human Genetics Annual Meeting.

- 2023 Travel Award, PGC Trainers Development Program and World Congress of Psychiatric Genetics.
- 2023 Best PhD Project, poster category, XXII Congress of the Graduate Program in Functional and Structural Biology.

RESEARCH FUNDING

- 01/2025–01/2026 **FAPESP 2024/11033-1 · Individual International Predoctoral Fellowship**
The role of copy-number variants in suicide attempt and suicidal ideation.
Principal investigator · Mentor: Niamh Mullins
- 03/2024–02/2029 **FAPESP 2023/05560-6 · Research grant**
Genetic and epigenetic approaches as predictive models in mental disorders.
Genetics analyst · Principal investigator: Marcos Leite Santoro
- 06/2023–02/2028 **FAPESP 2022/15880-5 · Individual Predoctoral Fellowship**
Impact of genetic variants on structural neuroimaging: a longitudinal approach in psychiatric disorders.
Principal investigator · Mentor: Marcos Leite Santoro
- 06/2019–07/2020 **FAPESP 2019/07629-8 · Undergraduate Research Fellowship**
Complete sequencing of cysteine-rich toxins from *Nephilingis cruentata* spider venom.
Principal investigator · Mentor: Alexandre Keiji Tashima

PUBLICATIONS (18 TOTAL; 2 FIRST-AUTHOR)

*Co-first authors.

Peer-reviewed articles

- Brañas MJA*, Ito LT*, Croci MS, Salto ABR, Doretto VF, Santoro ML, Ota VK, Rohde LA, Salum GA, Choi-Kain LW, Belangero S, Miguel EC, Hoffmann MS, Pan PM. Polygenic Risk for Major Depression: Diagnostic Specificity and Developmental Trajectories in an Admixed Youth Cohort. *Biological Psychiatry*. 2026. doi:10.1016/j.biopsych.2026.07.006.
- Damiano RF, Low DM, Ito LT, Santoro ML, Belangero S, Pan PM, Casella C, Schafer JL, Uher R, Szatmari P, Bryan CJ, Blumberg HP, Miguel EC, Rohde LA, Salum GA. Developmental predictors of suicide attempts from childhood to early adulthood: a 15-year prospective cohort study. *The Lancet Regional Health – Americas*. 2026;61:101531. doi:10.1016/j.lana.2026.101531.
- Legaz A, ... Ito LT, et al. The exposome of brain aging across 34 countries. *Nature Medicine*. 2026;32:1838–1851. doi:10.1038/s41591-026-04302-z.
- Fonseca L, Ito LT, Ziebold C, Rohde LA, Miguel EC, Bressan RA, Pan PM, Belangero SI, Santoro ML, Gomes FV, Grace AA, Gadelha A. Executive function mediates the effects of genetic liability to schizophrenia on behavior and functioning in a community sample of children and adolescents. *Schizophrenia Research*. 2026;290:9–16. doi:10.1016/j.schres.2026.01.014.
- Cabrera-Mendoza B, Panzenhagen AC, de Anda-Jáuregui G, He J, Magalhaes PVS, Martínez-Magaña JJ, Martínez-Levy GA, Qiu D, Sotelo D, Torres D, Medrano JV, Vidal G, Velásquez MM, Nicolini H, Polimanti R, Kaster MP, Docherty AR, Kimbrel NA, Genis-Mendoza AD, Ito LT, Shansis FM, Rovaris DL, Folego-Temoteo I, Fries GR, Suicide Working Group from the Latin American Genomics Consortium. Genomic Studies of Suicidal Thoughts and Behaviors in Latin American Populations: A Systematic Review. *Biological Psychiatry*. 2026. doi:10.1016/j.biopsych.2025.12.009.
- Salum GA, Giusti C, Souza L, Juk J, Costa R, Sugaya L, Caye A, Simioni A, Rocha PB, Manfro GG, Ito LT, Silva-Jr F, Duarte I, Esper N, Anés M, Damiano RF, Satterthwaite TD, Carvalho CM, Bado P, Hoffmann MS, Schafer JL, Casella CB, Evans-Lacko S, Ziebold C, Rocha R, Zugman A, Jackowski A, Gadelha A, Hoexter MQ, Madruga C, Grassi-Oliveira R, Cataneo A, Tyrka A, Paus T, Pine DS, Leibenluft E, Stringaris A, Merikangas KR, Milham MP, Franco A, Santoro M, Sato JR, Ota V, Polanczyk GV, Mari JJ, Bressan RA, Miguel EC, Rohde LA, Belangero S, Pan PM. Cohort Profile: Brazilian High-Risk Cohort for Mental Health Conditions (BHRC). *International Journal of Epidemiology*. 2025. doi:10.1093/ije/dyaf192.
- Arendt JA, Zamariolli M, Almodobar L, Ito LT, Ormond R, Oliveira AM, Ota V, Rohde LA, Miguel EC, Pan PM, Bressan RA, Salum GA, Belangero SI, Santoro ML. Copy Number Variations in the Brazilian High-Risk Cohort for Mental Conditions. *Brazilian Journal of Psychiatry*. 2025. doi:10.47626/1516-4446-2024-3765.
- Doretto VF, Salto ABR, Schivoletto S, Zugman A, Oliveira MC, Brañas MJAA, Croci MS, Ito LT, Santoro ML, Jackowski AP, Bressan RA, Rohde LA, Salum GA, Miguel EC, Pan PM. Childhood Maltreatment and the Structural Development of Hippocampus across Childhood and Adolescence. *Psychological Medicine*. 2025. doi:10.1017/S0033291724001636.
- Duarte I, Hoffmann MS, Salum GA, Leffa DT, Belangero S, Santoro ML, Ota V, Ito LT, Pan PM, Farhat LC, Murray AL, Miguel EC, Kieling C, Rohde LA, Caye A. Enhanced neurobiological biomarker differentiation for attention-deficit/hyperactivity disorder through a risk-informed design. *European Child & Adolescent Psychiatry*. 2024. doi:10.1007/s00787-024-02622-4.
- Salto ABR, Salum GA, Hoffmann MS, Santoro ML, Zugman A, Pan PM, Belangero SI, Ito LT, Doretto VF, Croci MS, Brañas MJAA, Giusti C, Silva-Jr FD, Ribeiro SM, Miguel EC, Leckman JF. The trajectory of anxiety symptoms during the transition from childhood to young adulthood is predicted by IQ and sex, but not polygenic risk scores. *JCPP Advances*. 2024. doi:10.1002/jcv2.12268.

- Bandeira CE, Grevet EH, Vitola ES, Silva BS, Cupertino RB, Picon FA, **Ito LT**, Tavares MEA, Rovaris DL, Grimm O, Bau CHD. Exploring Neuroimaging Association Scores in adulthood ADHD and middle-age trajectories. *Journal of Psychiatric Research*. 2024. doi:10.1016/j.jpsychires.2024.06.025.
- **Ito LT**, Miyamoto JG, Sant'Anna SS, Grego KF, Tanaka-Azevedo AM, Tashima AK. Unveiling the peptidome diversity of Lachesis muta snake venom: Discovery of novel fragments of metalloproteinase, L-amino-acid oxidase, and bradykinin-potentiating peptides. *Biochemical and Biophysical Research Communications*. 2023. doi:10.1016/j.bbrc.2023.10.022.
- Honorato-Mauer J, Xavier G, Ota VK, Chehimi SN, Mafra F, Cuóco C, **Ito LT**, Ormond R, Asprino PF, Oliveira A, Bugiga AVG, Torrecilhas AC, Bressan RA, Manfro GG, Miguel EC, Rohde LA, Pan PM, Salum GA, Pellegrino R, Belangero SI, Santoro ML. Alterations in microRNA of extracellular vesicles associated with major depression, attention-deficit/hyperactivity and anxiety disorders in adolescents. *Translational Psychiatry*. 2023. doi:10.1038/s41398-023-02326-4.

Preprints

- Colbert SMC, ... **Ito LT**, et al. Genome-wide association studies identify 77 loci for suicidality and provide novel biological insights. *medRxiv*. 2025. doi:10.1101/2025.10.22.25338076.
- Engchuan W, Shanta O, ... **Ito LT**, et al. Psychiatric disorders converge on common pathways but diverge in cellular context, spatial distribution, and directionality of genetic effects. *medRxiv*. 2025. doi:10.1101/2025.07.11.25331381.
- Shanta O, Klein M, ... **Ito LT**, et al. A cross-disorder analysis of CNVs finds novel loci and dose-dependent relationships of genes to psychiatric traits. *medRxiv*. 2025. doi:10.1101/2025.07.11.25331310.
- Baltramonaityte V, ... **Ito LT**, et al. Shared genetic architecture of brain age gap across 30 cohorts worldwide. *medRxiv*. 2025. doi:10.64898/2025.12.23.25342890.
- Amaral JV, Jobim GS, Pacheco JPG, **Ito LT**, Ormond R, Santoro ML, Belangero SI, Gadelha A, Miguel EC, Bressan RA, Pan PM, Rohde LA, Salum GA, Hoffmann MS. Clinical utility of loneliness as a symptom of major depression in youth. *PsyArXiv*. 2025. doi:10.31234/osf.io/24kwe.

SELECTED PRESENTATIONS

2025	Large-Scale Analysis of Copy Number Variants in the Etiology of Suicide Attempt and Suicidal Ideation. Oral presentation, World Congress of Psychiatric Genetics, Cancún, Mexico.
2025	Investigating the role of copy-number variants in suicide attempt and suicidal ideation in the Psychiatric Genomics Consortium. Poster, American Society of Human Genetics Annual Meeting, Boston, United States.
2024	Neuroimaging quality control in the Brazilian High-Risk Cohort Study: Progress and methods. Oral presentation, 3rd Meeting of the Center for Innovation and Research in Mental Health, São Paulo, Brazil.
2023	Impact of genetic variants on structural neuroimaging: a longitudinal approach in mental disorders. Poster, XXII Congress of the Graduate Program in Functional and Structural Biology, São Paulo, Brazil.
2023	Polygenic risk score and its relationship with multiple psychiatric disorders in the Brazilian population. Poster, World Congress of Psychiatric Genetics, Montreal, Canada.
2022	Polygenic score in the Brazilian High-Risk Cohort Study. Oral presentation, 1st Meeting of the Center for Innovation and Research in Mental Health, São Paulo, Brazil.

TEACHING, SERVICE, AND LEADERSHIP

2023–2024	Organizing Committee of Events, Graduate Program in Functional and Structural Biology, UNIFESP.
2020–2022	Volunteer tutor in Mathematics and Biostatistics using R, UNIFESP.
2020–2022	Organizing Committee, Journey of Biomedicine UNIFESP.
2022	Organizing Committee, Biomed Experience UNIFESP.
2020	Student Representative, Campus São Paulo Council, UNIFESP.
2019–2021	Financial Director, Leal Prado Academic Center, UNIFESP.